

CY 401: Introduction to Quantum Chemistry & Molecular Spectroscopy

Course Code	CY 401
Course Name	Introduction to Quantum Chemistry & Molecular Spectroscopy
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	BS Chemical Sciences
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Module I:** Introduction to Schrödinger equation, Bohr's atom, De Broglie's Wave, wave-particle duality, Light-atom/molecule interaction, Introduction to optical spectroscopy, time- independent Schrodinger equation, Particle in a box, Quantum Mechanics of Hydrogen Atom. (20 Hours)
- **Module II:** Heisenberg's Uncertainty Relation, Operators, Commutators, Eigenvalues and Eigenvectors, absorption and emission spectra, Boltzmann Energy distribution, Principle of equipartition of energy Einstein's Semiclassical model, Born Oppenheimer Approximation, Beer-Lambert Law. (14 Hours)
- **Module III:** Diatomic Vibrational Spectra: Harmonic Model, Morse Oscillator Model, Molecular Vibrations in Polyatomic Molecules, Diatomic rotational spectra, rotation of polyatomic molecules, electronic absorption, and emission spectra. (8 Hours)

Text Books

1. Peter Atkins, Julio de Paula, James Keeler, **Physical Chemistry**, Oxford University Press, 2018.

References

1. Thomas Engel, **Quantum Chemistry & Spectroscopy**, 3rd Edition, Pearson, 2015.
2. Donald A. McQuarrie & John D. Simons, **Physical Chemistry: A molecular approach**, Indian Reprint, Viva books, 2019.